

Sahil Wadhvani

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Education

University of Southern California, MS in Computer Science

Jan 2025 – Dec 2026 (*Expected*)

- GPA: 3.71 / 4.0

- **Coursework:** Distributed Systems, Analysis of Algorithms, Software Engineering for Security, Web Technologies

Pune University, Bachelor of Engineering in Computer Science

Jun 2020 – May 2024

- GPA: 9.04 / 10.0

- **Coursework:** Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming

Experience

Software Engineer Intern, Microsoft – Redmond, WA

May 2026 – Aug 2026

- Achieved **94.4% run reliability** across **6 production regions** by architecting a **globally distributed telemetry pipeline** using **Python** and Azure Data Explorer to process **6.4M+ daily device records**.
- Corrected reported **P95 onboarding latency from hours to minutes** by developing a **root-cause classification engine** in KQL that analyzed **76.5M records** to isolate system delays and define **onboarding SLOs**.
- Reduced **dashboard load times to sub-second** and extended historical data tracking from **30 to 365 days** by pre-computing data aggregates, delivering **3 self-service Grafana dashboards** that surfaced **2 critical production defects**.

Software Engineer Intern, nowfluence – Los Angeles, CA

Feb 2026 – May 2026

- **Owned and deployed the backend** for an AI-assisted influencer discovery platform, building a scalable **information retrieval system** with **Django, PostgreSQL, and AWS ECS** to improve search relevance and retrieval quality.
- Improved **API responsiveness and fault tolerance** under load by implementing **multi-layer Redis** caching and **asynchronous Celery workflows**, integrating OpenAI LLMs with **deterministic fallbacks** for reliable execution.

Associate Software Developer, Coditas – Pune, India

Jan 2024 – Dec 2024

- Processed **10,000+ daily transactions** by engineering **Go and PostgreSQL backend services** for a **latency-sensitive stock trading platform** and consolidating fragmented microservices into a unified architecture.
- Reduced downstream analytics **query latency by 25%** by optimizing PostgreSQL queries, introducing **Redis caching**, and parallelizing execution with **Go goroutines**.

Software Development Intern (Security), EzConverse Media – India

Aug 2023 – Nov 2023

- Reduced manual incident reporting and **triage time by 30%** across internal services by engineering a **full-stack Vulnerability Management System** (React.js, Node.js, MongoDB), **automating remediation workflows** through secure REST APIs.
- **Identified and remediated 10+ critical vulnerabilities** through **threat modeling and security testing** with Burp Suite and Metasploit, hardening APIs with **RBAC, JWT authentication**, input validation, and audit logging.

Skills & Certifications

Programming Languages: Python, C/C++, Java, Go, TypeScript, JavaScript, SQL, Bash

Systems & Fundamentals: Distributed Systems, Concurrency, Multithreading, System Design, Networking, Unix/Linux

Backend, Web & Data: Django, FastAPI, Node.js, React.js, Next.js, REST APIs, PostgreSQL, MongoDB, Redis, Celery

AI/ML & Information Retrieval: Machine Learning, PyTorch, BERT, Natural Language Processing, Large Language Models, Pandas

Cloud, Security & Developer Tools: GCP, Azure, AWS, Docker, Git, Copilot, Antigravity, Threat Modeling, CEH v12, CCNA

Technical Projects

SentinelX – Full-Stack Security Operations Platform

[GitHub](#)

- Built a **full-stack security operations platform** with Next.js 14, TypeScript, FastAPI, and PostgreSQL, centralizing **event ingestion, investigation, and response workflows** in a single analyst interface.
- Implemented **least-privilege access** and **real-time response workflows** using custom **RBAC, JWT-based sessions**, and React Query state synchronization, enabling analysts to securely investigate and act on system events.

AI Trading Agent – Autonomous Fintech Backend

[GitHub](#)

- Developed a **highly concurrent Python backend** utilizing local LLMs (Ollama) and SQLite to autonomously parse real-time financial sentiment, scanning over **96,000 data streams** continuously.
- **Constrained autonomous execution** by engineering a **custom orchestration layer** with **10 deterministic risk gates** before trade execution, separating probabilistic LLM reasoning from safety-critical decision logic.